

E-In Son

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RESEARCH INTERESTS

My research focuses on learning-based navigation for robots in real-world environments, spanning traversability estimation in unstructured terrain, visual navigation, and world models. Going forward, I aim to develop world models and navigation policies that generalize across environments and robot embodiments.

EDUCATION

Seoul National University

Mar 2021 – Aug 2027 (Expected)

Ph.D. Student, Dept. of Electrical & Computer Engineering

Seoul, Korea

Vehicle Intelligence Laboratory. (Advisor: Prof. Seung-Woo Seo)

Konkuk University

Mar 2015 – Feb 2021

B.S., Electrical & Electronics Engineering

Seoul, Korea

Summa Cum Laude (GPA 4.33 / 4.5).

PUBLICATIONS

[1] **Feeling Terrain Before Crossing: World Models for Off-Road Navigation.**

E-In Son, Dong-Wook Kim, Ji-Hoon Hwang, Kangsun Lee, Jisung Bae, Jung-Taak Kim, Seung-Woo Seo.

Under review, 2026. [\[Paper\]](#)

[2] **Which Terrain Is Better? Preference Learning with VLM Prototypes for Off-Road Traversability Ranking.**

Ji-Hoon Hwang, Jisung Bae, E-In Son, Dong-Wook Kim, Jung-Taak Kim, Seung-Woo Seo.

Under review, 2026. [\[Paper\]](#)

[3] **CanonNav: Disentangling Navigation Behavior from Camera Geometry in Cross-Platform Visual Navigation.**

Dong-Wook Kim, Ji-Hoon Hwang, E-In Son, Mintaek Oh, Seung-Woo Seo.

Under review, 2026. [\[Paper\]](#)

[4] **Ordinal Neural Collapse as a Representation Prior for Visual Navigation.**

E-In Son^{*}, Jung-Taak Kim^{*}, Seung-Woo Seo.

European Conference on Computer Vision (ECCV), 2026. [\[Paper\]](#) [\[Project\]](#)

[5] **PTS-Map: Probabilistic Terrain State Map for Uncertainty-Aware Traversability Mapping in Unstructured Environments.**

Dong-Wook Kim, E-In Son, Chan Kim, Ji-Hoon Hwang, Seung-Woo Seo.

IEEE Robotics and Automation Letters (RA-L), 2025. [\[Paper\]](#) [\[Code\]](#)

[6] **Follow the Footprints: Self-supervised Traversability Estimation for Off-road Vehicle Navigation based on Geometric and Visual Cues.**

Yurim Jeon, E-In Son, Seung-Woo Seo.

IEEE International Conference on Robotics and Automation (ICRA), 2024. [\[Paper\]](#) [\[Code\]](#)

[7] **Adaptive Robot Traversability Estimation Based on Self-Supervised Online Continual Learning in Unstructured Environments.**

Hyung-Suk Yoon, Ji-Hoon Hwang, Chan Kim, E-In Son, Se-Wook Yoo, Seung-Woo Seo.

IEEE Robotics and Automation Letters (RA-L), 2024. [\[Paper\]](#)

- [8] **Traversability-aware Adaptive Optimization for Path Planning and Control in Mountainous Terrain.**
Se-Wook Yoo*, [E-In Son*](#), Seung-Woo Seo.
IEEE Robotics and Automation Letters (RA-L), 2024. [\[Paper\]](#) [\[Project\]](#)

AWARDS & HONORS

Earth Rovers Challenge (1st Place) IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) <i>Real-world autonomous navigation competition, AI policies vs. human teleoperators.</i>	Oct 2024
Academic Excellence Scholarship Konkuk University	2018 – 2020

RESEARCH PROJECTS

Challengeable Future Defense Technology R&D Program (Phase 2) Agency for Defense Development (ADD) <i>Leading the planning stack, from learning-based path prediction to real-time safe replanning, for an end-to-end visual navigation system operating in unknown, hazardous terrain.</i>	2024 – 2026
Challengeable Future Defense Technology R&D Program (Phase 1) Agency for Defense Development (ADD) <i>Developed the path tracking module for accurate and stable trajectory following.</i>	2022 – 2023

TEACHING EXPERIENCE

Seoul National University , Teaching Assistant
SNU in Silicon Valley Program, joint with UC Berkeley (<i>Summer 2024</i>)
Fundamentals of Random Variables and Probability Processes (<i>Spring 2023</i>)
Network Protocol and Design (<i>Fall 2021, 2023</i>)

ACADEMIC SERVICE

Conference Reviewer: ICRA, IROS, ECCV
Journal Reviewer: RA-L

SKILLS

Tools and Languages	Python, C/C++, PyTorch, ROS
Communication	Korean (native), English

REFERENCE

Advisor: Prof. Seung-Woo Seo sseo@snu.ac.kr [Website](#)